

AI First Development with GitHub Copilot

Duration:

20 Hours (3 Hours/Day)

Course Overview:

This specialization empowers software developers, engineers, testers and technology leaders to harness the full potential of **AI-assisted software development** using **GitHub Copilot** and related tools. The course blends hands-on coding practice with strategic understanding of how AI is transforming software engineering — from code creation to testing, documentation, and DevOps automation.

Participants will learn to **leverage GitHub Copilot and complementary AI tools** across the Software Development Lifecycle (SDLC), implement **responsible AI practices**, and prepare for the **next generation of AI-augmented development environments**.

Course Objectives:

Upon completion, participants will be able to:

- Apply AI-assisted coding, debugging, and refactoring techniques using GitHub Copilot.
- Integrate AI tools into daily workflows for faster, cleaner, and more maintainable code. Automate documentation, testing, and DevOps with AI-enabled extensions.
- Understand and apply governance, security, and ethical frameworks for AI-generated code.
- Evaluate and deploy AI augmentation strategies within enterprise software ecosystems.

Pre-requisites:

- Basic programming experience in any modern language (e.g., Python, JavaScript, C#, Java).
- Familiarity with GitHub repositories, version control, and IDEs (VS Code preferred).
- Optional: Exposure to Agile or DevOps workflows.

Course Outline:

Module 1: Foundations of AI-Augmented Development (Day 01)

- The evolution of AI in software engineering
- AI's role in accelerating developer productivity
- Introduction to Gen AI, LLM, core concepts
- AI assistant landscape:
 - Chat-based assistants
 - IDE-based coding assistants
 - CLI-based assistants
- Where AI fits in modern engineering workflows
- Safe, ethical, and responsible AI usage
- Enterprise considerations for AI adoption

Outcome:

Participants share a common mental model of AI, its capabilities, limits, and role in professional software delivery.

Module 2: Prompting Fundamentals & AI Interaction Patterns (Day 02)

- How AI interprets intent and context
- Prompt anatomy: intent, context, constraints, output
- Zero-shot, one-shot, few-shot prompting, CoT
- Iterative prompting and refinement loops
- Conversational vs IDE-based prompting
- Overview of GitHub Copilot within GitHub and GitHub Enterprise environments
- Setting up Copilot-ready environments (VS Code, extensions, configuration)
- Model selection in AI coding assistants

Demo/Hands-on:

Configure GitHub Copilot settings in VS Code, validate authentication and repository access, and generate initial AI-assisted code suggestions for a simple business use case

Outcome:

Participants can consistently guide AI systems to produce relevant, high-quality outputs.

Module 3: AI-Assisted Development Fundamentals (Day 03)

- AI-assisted development workflows
- How GitHub Copilot understands developer intent, prompts, and context windows
- Writing effective prompts for intelligent code completion using GitHub Copilot
- Ask-oriented and comment-driven prompts
- Code generation, explanation, and debugging
- Human-in-the-loop development
- Working across languages and frameworks with GitHub Copilot
- Real-time code suggestions, debugging, and optimization assistance from GitHub Copilot

Demo/Hands-on:

Build a simple end-to-end project using GitHub Copilot-assisted development, focusing on prompt refinement and contextual guidance..

Outcome:

Participants confidently use AI assistants as productivity accelerators while retaining control over design and correctness.

Module 4: Quality, Testing, and Maintainability with AI (Day 04)

- Using AI to improve code readability and documentation
- AI-assisted test generation and enhancement
- Understanding AI hallucinations and incomplete logic
- Validation strategies:
 - Tests
 - Reviews
 - Comparisons
- Refactoring and optimization with AI assistance
- Improving maintainability of existing codebases

Hands-on:

- Explore repetitive task automation using AI plugins/extensions: generating boilerplate, creating test scaffolds, or documenting code automatically.
- Refactor existing code using Copilot suggestions and optimise

Outcome:

- Participants learn to use AI to raise quality and reliability — not just speed.
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Module 5: End-to-End Development & Context Awareness (Day 05)

- AI as a pair programmer for building an app end-to-end
- AI as a pair programmer for larger codebases
- Navigating and understanding legacy systems with AI
- End-to-end development acceleration
- Managing complexity in multi-component applications
- Context limitations in AI-assisted development
- Introduction to controlled context sharing concepts

Hands-on:

- Use Copilot to build end-end-end application from the scratch
- Navigate a multi-file codebase, add features across components, and refactor interconnected logic
- Demonstrate where Copilot lacks business or organizational context and visually introduce the MCP concept

Outcome:

- Participants experience AI-assisted development at scale and understand why context management becomes critical.

Module 6: Integrating AI into the Software Development Lifecycle (Day 06)

- Applying AI across the software delivery lifecycle:
 - Requirements and understanding
 - Development and reviews
 - Testing and validation
- Secure and responsible prompting practices
- Governance considerations for AI-assisted workflows
- Introduction to Model Context Protocol (MCP):
 - Purpose and benefits
 - Controlled access to approved context
 - Enterprise readiness and policy alignment
- Embedding AI tools (GitHub Copilot) into Agile and DevOps workflows

Hands-on:

Create an AI-integrated SDLC workflow using GitHub Projects, Issues, and GitHub Copilot collaboration, including AI-assisted code reviews and test coverage improvement.

Outcome:

Participants can embed Copilot across the software lifecycle and measure its impact with prompt-driven metrics.

Module 6: Measuring Value, Automation & Sustainable Adoption (Day 07)

- Measuring AI impact on productivity and quality
- Identifying opportunities for AI-assisted automation
- Designing repeatable AI-enabled workflows
- Balancing efficiency, quality, and risk
- Adoption roadmap:
 - Individual
 - Team
 - Organization
- Managing AI coding assistants (including GitHub Copilot) in enterprise projects
- Integration with internal repositories, APIs, and enterprise access policies
- Policy, compliance, and organizational alignment (SOC 2, GDPR, HIPAA considerations)
- **Prompt governance:** defining rules for prompt usage, logging, and auditability
- **Best practices for enterprise enablement:**
 - User roles and access control
 - Repository and branch protection rules
 - Version control and review workflows
 - Secure data flow for AI-assisted development

Outcome:

Participants can articulate AI value, plan adoption, and apply AI-first practices beyond the training.

Participants learn how to implement GitHub Copilot securely within enterprise environments using governance policies, repository controls, and compliance-aligned SDLC practices.
